

Fenômenos de transporte $\Rightarrow$ Exercício manual

4.1) $\dot{Q} = h \cdot A \cdot \Delta T$; $d = 2,6 \text{ cm} = 0,025 \text{ m}$; $L = 0,15 \text{ m}$; $\Delta T = 244$; $h = 15 \text{ W/m}^2 \cdot \text{K}$

$$A = \pi dL + 2\pi \left(\frac{d}{2}\right)^2 = \pi (0,025)(0,15) + 2\pi (0,0125)^2 = 0,01276 \text{ m}^2$$

$$\dot{Q} = 15 \cdot 0,01276 \cdot 244 = 46,7016$$

4.2) $h = 29 \text{ W/m}^2 \cdot \text{K}$ [vento]; $Q = 700 \text{ W/m}^2 \cdot \text{K}^4$

$$T_{\infty} = 303 \text{ K}; T_s = T_{\infty} + \frac{Q}{h} = 303 + \frac{700}{29} = 303 + 24,14$$

$$T_s = 303 + 24,14 = 327,14 \text{ K} = 54^\circ \text{C},,$$

4.3) $A = 30 \text{ m}^2$; $L = 0,36$; $T_{\text{int}} = 25^\circ \text{C}$; $T_{\text{ext}} = 0^\circ \text{C}$; $K = 0,72 \text{ W/m} \cdot \text{K}$

$$\dot{Q} = \frac{KA\Delta T}{L} = \frac{0,72 \cdot 30 \cdot 25}{0,3} = \frac{540}{0,3} = 1800 \text{ W}$$